

External Validation Atlas · 15+ cohorts confirming the DM1/DM2 axis across bulk, methylation, single-cell, protein, drug, and pan-cancer layers

Direction-consistent across all 14/15 external cohorts (GSE286332 is the predicted HT-route attenuation per the two-axis model)

Bulk RNA

n = 504

TCGA-THCA primary

Discovery

DM1 prevalence 28.4 %

anchor cohort

Bulk RNA

n = 632

Lee GSE213647

Cohen's d (DM1–DM2)

d = +0.95

p < 10^{−15}

Bulk RNA

n = 260

K2 PRJEB11591

% DM1-like

37.8 % (calibration caveat)

K-S preserved

Bulk RNA

n = 117

MSK-IMPACT advanced

Cox HR (DM1 vs DM2)

HR = 2.67 [1.17, 6.10]

p = 0.018

Bulk RNA

n = 18

GSE286332 PTC+HT

MAPK × Panel ρ

ρ = −0.040 (HT route)

NS (predicted)

Bulk RNA

n = 37

GSE76039 PDTC+ATC

Convergence

5 / 8 panel overlap

Landa silenced list

Bulk RNA

n = 28

GSE126698 Korean PTC

MAPK × Panel ρ

ρ = −0.18

trend

Methylation

n = 503

HM450 TCGA

TPO d (DM1 vs DM2)

d = 2.30

p = 1.9 × 10^{−18}

Single-cell

n = 6

Pu 2021 GSE184362

Per-patient r

r = 0.798 - 0.886

p < 10^{−10} Bonferroni

Single-cell

n = 23

Lu 2023 GSE193581

Spearman r

r = 0.86 median

p < 10^{−10}

Single-cell

n = 14

GSE241184 Phase 1

Author-independence

r = 0.79

concordant

Single-cell

n = 11

GSE232237 External

Pseudobulk d

d = +1.65

p = 4 × 10^{−9}

Protein

n = 336

Mun 2025 proteome

Thyroid-diff d

d = −1.91

7 / 7 sign-consistent

Bulk RNA

n = 52

GSE151179 post-RAI

Pre vs post-RAI d

d = −1.01

p = 1 × 10^{−4}

Drug

n ≈ 766

DepMap + PRISM

AZD-0364 MEK d

d = −0.594

FDR = 5.9 × 10^{−7}

Pan-cancer

n = 11,069

TCGA pan-cancer

LGG / LUAD HR

HR 44.7 / 19.0

p = 1.3 × 10^{−5} / 8.6 × 10^{−4}